DIC19
Supply IR Drop and Decoupling

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Rebuild the Lab 19 supply IR-drop and decoupling comparison from PDF pages 78–80, including the peak-current explanation and capacitor costs.

2 Specification Mapping

S denotes simulated checks; D denotes analytical discussion.

Requirement and PDF pages	Deck / check	Status
Fig.19.1 four stages and two 1-ohm resistors (78,80)	D01/C0	S
Observe external/internal rails and R1/R2 currents (78,80)	D01; printed traces	S
10pF/100pF/1nF comparison (79–80)	D01/C1–3	S
Peak current, costs and use cases (80)	Results/discussion	S+D

3 Circuit and Design Derivation

Four inverter stages have multipliers 30,60,180,540 with unit $W_p/W_n/L = 3.6/1.8/0.25\,\mu$m. A 1 Ω resistor connects each external supply/return to its internal rail. The final stage additionally drives 10 fF to internal ground. Decoupling connects internal VDD to internal ground; it is not attached across the ideal external source. 10^{-21} F approximates no added capacitor.

$$\Delta V_R = IR, \qquad i_C = C\, d(V_{DD,int} - V_{GND,int})/dt.$$

This is a lumped-RC supply model without extracted metal or package inductance.

4 Simulation Setup

The model is supplied separately on the user's computer. These are 1 V, 27 °C teaching conditions; the handout's old-process 2.5 V numbers are not pass thresholds. Actual PrimeSim HSPICE S-2021.09 runs use `accurate`, `runlvl=5` and `.print/.measure` data. Every `.alter` resets baseline

parameters and temperature. All cases completed without errors or failed measurements; model and convergence notices may remain. Public netlists retain only `* MODEL_SETUP_REQUIRED`, with no model path. Full data are under the lab's `results/`; large files are split by deck and indexed by `simulation_data.json`.

One deck contains four cases. The 0–1 V input uses 50 ps edges, 1 ns delay, 2 ns high time and a 4 ns period. Simulation lasts 20 ns with at most 2 ps steps. All summaries use the same 4–20 ns observation window.

5 Results and Specification Checks

C	$V_{rail,min}$	$\Delta V_{rail,pp}$	$I_{R1,pk}$	$I_{R2,pk}$	P_{avg}
0	0.93453	63.350	32.734	32.738	9.923
10 pF	0.93486	62.301	32.570	32.573	9.930
100 pF	0.94510	50.913	27.447	27.449	9.975
1 nF	0.96779	23.425	17.507	17.283	9.520

Minimum rail voltage is in volts; rail peak-to-peak ripple in mV; currents in mA; power in mW. Minimum differential supply rises from 0.93453 V without decoupling to 0.96779 V at 1 nF. R1 peak current falls from 32.734 to 17.507 mA, with a similar R2 trend. The small improvement at 10 pF shows that capacitance must be evaluated relative to switched charge.

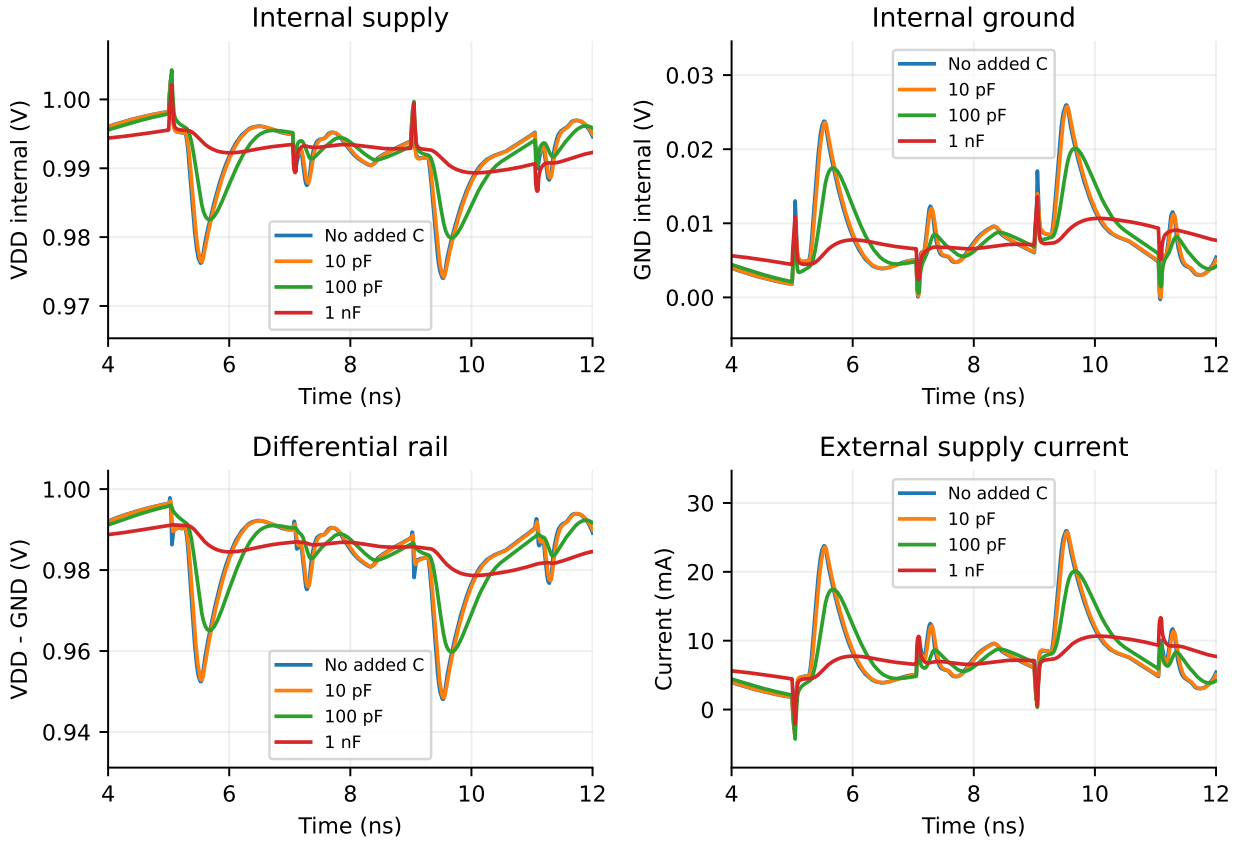


Figure 1: Actual internal supply, ground, differential rail and source current. The figure zooms to 4–12 ns; summaries use all of 4–20 ns.

6 Discussion

During rapid switching, the capacitor supplies or absorbs local charge and smooths current through the external resistors, reducing peak voltage drop. It does not remove average DC IR drop. Larger capacitance improves transient supply but consumes physical area and increases charging/startup time; real devices also have leakage, ESR/ESL and fabrication constraints. No physical area is inferred from an ideal capacitor value.

Decoupling need depends on allowable noise, simultaneous switching and supply impedance. Improving supply routing or staggering switching can also help; local fast current loads motivate local capacitance. Finite-window average power must not be equated with peak-current reduction: capacitor charging and initial conditions also affect the average.

7 Conclusion

All four capacitor conditions and both supply branches were simulated. A 1 nF capacitor substantially reduces differential rail ripple, without making the rails perfectly fixed or providing physical power-integrity signoff.

8 MOS transistor schematic

These editable vector schematics connect equal node names within a circuit. Primitive-cell names are local pins; the hierarchy defines their connections. B denotes the body terminal; 0 is global ground. Each panel identifies its netlist and scope; measurement sources are connected as described. The actual inverter cell and complete supply/chain connection of `01_ir_drop_decoupling.sp`.

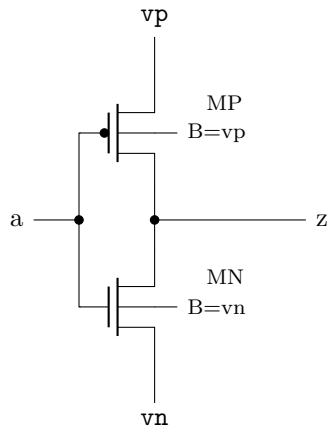


Figure 2: Exact inv cell used by X0–X3.

The cell uses $W_P = 3.6k\,\mu$m, $W_N = 1.8k\,\mu$m, $L = 0.25\,\mu$m. Both its sources and bodies use the local vp/vn rails.

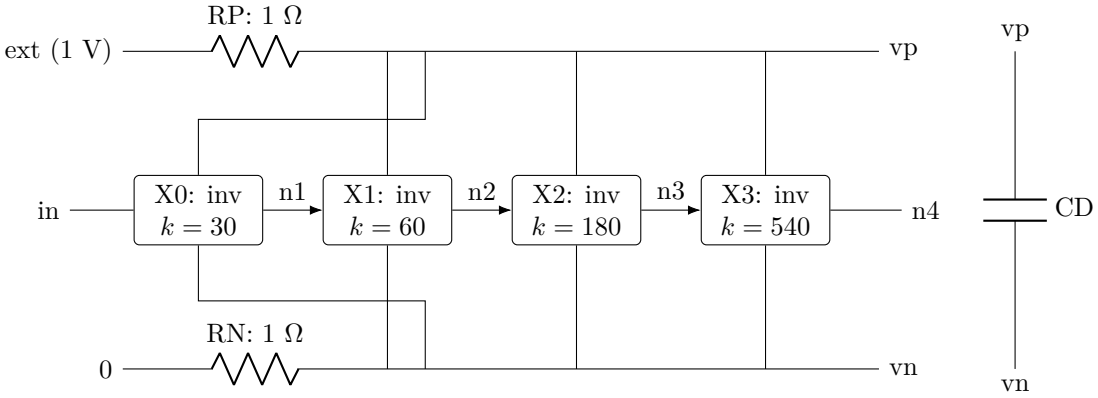


Figure 3: Four-stage chain, rail resistance and the decoupling branch.

CD=cd is between vp and vn, with cases approximately 0, 10 pF, 100 pF and 1 nF. The output capacitor CL=10 fF connects n4 to vn. VIN drives in relative to global 0. The blocks expand exactly to the two-MOS cell above; k applies to both transistor widths.